Dr Fan Zhang

Contact Information

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Research Interests

General Robot Manipulation, Generalizable Representation Learning (Sim-to-Real Transfer, Self-Supervision); Generative Models (World Action Models, Flow-based Policies)

Professional Appointments

Senior Scientist, 2025-present
Guest Scientist, 2024
Honda Research Institute EU

- Directing a specialized team focused on initial general dexterous manipulation research
- Accepted a Research Scientist role at Meta FAIR, 2025 (impacted by restructuring)

Eric and Wendy Schmidt AI in Science Postdoctoral Fellow, Schmidt Futures, 2023-2024
Research Associate, 2021-2023
Imperial College London (UK Global Talent Visa, sponsored by Royal Academy of Engineering)

Education

Ph.D. in Electrical and Electronic Engineering (Robotics), 2016-2020
Imperial College London
Thesis: Perception and Manipulation in Robotic-Assisted Dressing
Supervisor: Prof. Yiannis Demiris

Awards

Outstanding Paper Award, IROS 2025 Workshop on The Art of Robustness	2025
Best Research Paper (ECR), AI & Robotics Research Awards, held by UKRI TAS Hub	2025
The UK Best PhD in Robotics Award 1st place, held by Advanced Robotics @ Queen Mary	2020
Best Student Paper Award, IEEE International Conference on Mechatronics and Automation	2016

Highlighted Publications

Learning Robot Manipulation from Audio World Models
Fan Zhang, Michael Gienger
arXiv, 2025. (paper)

Affordance-based Manipulation with Flow Matching
Fan Zhang, Michael Gienger
IEEE Transactions on Robotics (conditionally accepted), 2026. (paper, website, code 240+ ★)

Learning Garment Manipulation Policies towards Robot-Assisted Dressing
Fan Zhang, Yiannis Demiris
Science Robotics, 2022. (paper, website, code)

Probabilistic Real-Time User Posture Tracking for Personalized Robot-Assisted Dressing
Fan Zhang, Antone Cully, Yiannis Demiris
IEEE Transactions on Robotics, 2019. (paper, website, code)

Talks

Intelligent Systems and Networks (ISN) Group Seminar, Imperial College London	2025
IROS Workshop on Manipulation of Deformable Objects	2025
IPAB visitor seminar, Edinburgh Centre for Robotics	2025
YorRobots Seminar, University of York	2025
Autonomous Agents Research Group, The University of Edinburgh	2025
Adaptive and Intelligent Robotics Lab, Imperial College London	2025
TAMS, University of Hamburg	2024
Social AI & Robotics Laboratory, King's College London	2023
Tsinghua University, (live audience: 150,000)	2022
Apple	2022
Human Motion Analysis for Healthcare Applications, IET	2019
The Hamlyn Centre, Imperial College London	2017

In the Press

Robotic nurse can dress a mannequin in a hospital gown, New Scientist	2022
Robotic nurse that helps you dress could aid staff shortage, Bloomberg	2019
The Times, Daily Mail, Telegraph, TechXplore	2019

Academic Activities

Inaugural Meeting - Centre for AI in Assistive Autonomy	Invited Guest
IROS 2025 Session: Factory Automation and Failure Detection	Co-chair
RSS 2024 Workshop: Learning for Assistive Robotics	Organizer
Imperial-X Breaking Topics in AI conference, Schmidt Futures	Organizer
ICRA 2023 Workshop: Emerging Paradigms for Assistive Robotic Manipulation	Organizer
Frontiers in Robotics and AI-Robot Learning and Evolution	Review Editor
Scientific Reports, T-RO, IJRR, ICRA, IROS, RA-L, RSS, WACV, Humanoids	Reviewer
The Centre for AI in Assistive Autonomy, The University of Edinburgh	Collaborator
Doctoral annual review (Leonard Hinckeldey), The University of Edinburgh	Committee

Supervision & Mentorship

Leonard Hinckeldey (PhD, The University of Edinburgh, Multi-Agent RL)	2025-present
Jonathan Sutphen (Msc, TU Darmstadt, Robot Manipulation with Audio)	2025-2026
Amirreza Razmjoo (PhD, EPFL, Robot Failure Recovery)	2024-2025
Jenny Fu (PhD, Cornell University, Robot Character Learning)	2024-2025
Nikki Zhong (PhD, Imperial college London, Human Motion Modeling)	2021-2022

Research Experience

---- **General Multi-Modal Robot Manipulation**

- Multimodal world model for learning robot manipulation, with special focus on contrastive latent embedding and physics learning.
- Parameter-efficient prompt tuning for learning manipulation affordance with VLM.
- Generative learning for robot manipulation with special focus on flow matching policy (IEEE Transactions on Robotics, conditionally accepted).
- Applications: robot failure recovery with compositional diffusion models, robotic expressions and characters learning with flow matching, piano playing with audio world models (IROS, workshops at CoRL and ROMAN).
- Outstanding Paper Award at IROS 2025 Workshop on The Art of Robustness.
- Developed an open-source robot manipulation repository, achieving 240+ stars on GitHub (code).

---- Robot-Assisted Dressing for Bedridden Patients

- Deformable visual-tactile manipulation for garment grasping unfolding using deep Q learning from demonstration and model-based reinforcement learning (RA-L).
- Sim-to-real garment physics learning and robot manipulation policy transfer in physics domain using self-supervised learning (Science Robotics).
- Self-supervised learning for human motion modeling, including personalized user impairments modeling and dancing performance assessment (ICASSP).
- Building large real and synthetic dataset of garment, including RGB-D, event images and pointcloud.
- Real-time user posture tracking using vision and haptic information with a probabilistic particle filter (IEEE Transactions on Robotics).
- Building controllers for hierarchical multi-task human-robot interaction (ICRA, IROS).
- Best Research Paper (Early Career Researcher), 2025 AI & Robotics Research Awards, held by UKRI TAS Hub and Responsible Ai UK.
- UK Best PhD in Robotics Award 2020 1st place, held by the Centre for Advanced Robotics @ Queen Mary (ARQ).
- Live demos for National Health Service, Apple, etc.
- Media covered by Bloomberg, The Times, Daily Mail, Telegraph , NewScientist, TechXplore.
- The above works are supported by MURI UKRI Closed-Loop Multisensory Brain-Computer Interface for Enhanced Decision, TAS Trust Node UKRI Trustworthy Autonomous Systems Node in Trust, Innovate UK D-RISK Learning Edge Cases for Autonomous Vehicles.

Other Publications

Complete List (https://scholar.google.com/citations?user=X-OiekwAAAAJ&hl)

1) **Assistax: A Hardware-Accelerated Reinforcement Learning Benchmark for Assistive Robotics**
   Leonard Hinckeldey, Elliot Fosong, Rimvydas Rubavicius, Elle Miller, Trevor McInroe, Fan Zhang, Patricia Wollstadt, Stefano V. Albrecht, Subramanian Ramamoorthy
   arXiv, 2026. (paper)

2) **Composition of Conditional Diffusion Policies with Guided Sampling**
   Amirreza Razmjoo, Sylvain Calinon, Michael Gienger, Fan Zhang
   IROS, 2025. (paper, website)

3) **Generation of Real-time Robotic Emotional Expressions Learning from Human Demonstration in Mixed Reality**
   Chao Wang, Michael Gienger, Fan Zhang
   IROS 2025 Foundation Models for Robotic Design workshop. (paper, website)

4) **Contrastive Self-Supervised Learning for Automated Multi-Modal Dance Performance Assessment**
   Yun Zhong, Fan Zhang, Yiannis Demiris
   ICASSP, 2023. (paper, video, code)

5) **Visual-Tactile Learning of Garment Unfolding for Robot-Assisted Dressing**
   Fan Zhang, Yiannis Demiris
   RA-L, 2023. (paper, video)

6) **Learning Grasping Points for Garment Manipulation in Robot-Assisted Dressing**
   Fan Zhang, Yiannis Demiris
   ICRA, 2020. (paper, video)

7) **Personalized Robot-Assisted Dressing using User Modeling in Latent Spaces**
   Fan Zhang, Antone Cully, Yiannis Demiris
   IROS, 2017. (paper, video)